A Mechanistic Interpretation of Arithmetic Reasoning in Language Models using Causal Mediation Analysis

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Abstract

Mathematical reasoning in large language models (LMs) has garnered significant attention in recent work, but there is a limited understanding of how these models process and store information related to arithmetic tasks within their architecture. In order to improve our understanding of this aspect of language models, we present a mechanistic interpretation of Transformer-based LMs on arithmetic questions using a causal mediation analysis framework. By intervening on the activations of specific model components and measuring the resulting changes in predicted probabilities, we identify the subset of parameters responsible for specific predictions. This provides insights into how information related to arithmetic is processed by LMs. Our experimental results indicate that LMs process the input by transmitting the information relevant to the query from mid-sequence early layers to the final token using the attention mechanism. Then, this information is processed by a set of MLP modules, which generate result-related information that is incorporated into the residual stream. To assess the specificity of the observed activation dynamics, we compare the effects of different model components on arithmetic queries with other tasks, including number retrieval from prompts and factual knowledge questions.[1]

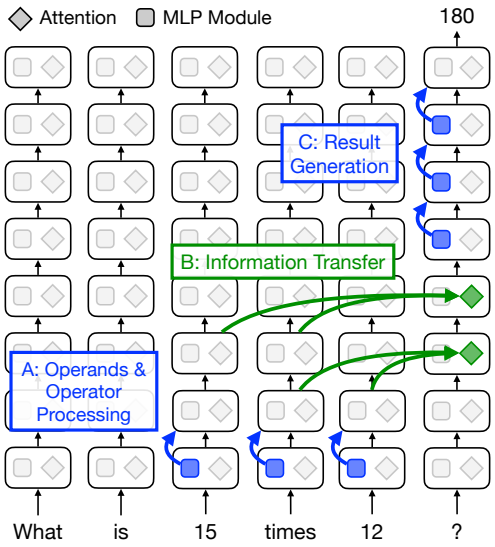

Figure 1: Visualization of our findings. We trace the flow of numerical information within Transformer-based LMs: given an input query, the model processes the representations of numbers and operators with early layers (A). Then, the relevant information is conveyed by the attention mechanism to the end of the input sequence (B). Here, it is processed by late MLP modules, which output result-related information into the residual stream (C).

1 Introduction

Mathematical reasoning with Transformer-based models (Vaswani et al., 2017) is challenging as it requires an understanding of the quantities and the mathematical concepts involved. While large language models (LMs) have recently achieved impressive performance on a set of math-based tasks (Wei et al., 2022a; Chowdhery et al., 2022; OpenAI, 2023), their behavior has been shown to be inconsistent and context-dependent (Bubeck et al., 2023). Recent literature shows a multitude of works proposing methods to improve the performance of large LMs on math benchmark datasets through enhanced pre-training (Spokoyny et al., 2022; Lewkowycz et al., 2022; Liu and Low, 2023) or specific prompting techniques (Wei et al., 2022b; Kojima et al., 2022; Yang et al., 2023, *inter alia*). However, there is a limited understanding of the inner workings of these models and how they store and process information to correctly perform math-based tasks. Insights into the mechanics behind LMs' reasoning are key to improvements such as inference-time correction of the model's behavior (Li et al., 2023) and safer deployment. Therefore, research in this direction is critical for the development of more faithful and accurate next-generation LM-based reasoning systems.

In this paper, we present a set of analyses aimed

[1] Our code and data is available at https://github.com/alestolfo/lm-arithmetic.

at mechanistically interpreting LMs on the task of answering simple arithmetic questions (e.g., *"What is the product of 11 and 17?"*). In particular, we hypothesize that the computations involved in reasoning about such arithmetic problems are carried out by a specific subset of the network. Then, we test this hypothesis by adopting a causal mediation analysis framework (Vig et al., 2020; Meng et al., 2022), where the model is seen as a causal graph going from inputs to outputs, and the model components (e.g., neurons or layers) are seen as mediators (Pearl, 2001). Within this framework, we assess the impact of a mediator on the observed output behavior by conducting controlled interventions on the activations of specific subsets of the model and examining the resulting changes in the probabilities assigned to different numerical predictions.

Through this experimental procedure, we track the flow of information within the model and identify the model components that encode information about the result of arithmetic queries. Our findings show that the model processes the input by conveying information about the operator and the operands from mid-sequence early layers to the final token using attention. At this location, the information is processed by a set of MLP modules, which output result-related information into the residual stream (shown in Figure 1). We verify this finding for bi- and tri-variate arithmetic queries across four pre-trained language models with different sizes: 2.8B, 6B, and 7B parameters. Finally, we compare the effect of different model components on answering arithmetic questions to two additional tasks: a synthetic task that involves retrieving a number from the prompt and answering questions related to factual knowledge. This comparison validates the *specificity* of the activation dynamics observed on arithmetic queries.

2 Related Work

Mechanistic Interpretability. The objective of mechanistic interpretability is to reverse engineer model computation into components, aiming to discover, comprehend, and validate the algorithms (called circuits in certain works) implemented by the model weights (Räuker et al., 2023). Early work in this area analyzed the activation values of single neurons when generating text using LSTMs (Karpathy et al., 2015). A multitude of studies have later focused on interpreting weights and intermediate representations in neural networks (Olah et al., 2017, 2018, 2020; Voss et al., 2021; Goh et al., 2021) and on how information is processed by Transformer-based (Vaswani et al., 2017) language models (Geva et al., 2021, 2022, 2023; Olsson et al., 2022; Nanda et al., 2023). Although not strictly mechanistic, other recent studies have analyzed the hidden representations and behavior of inner components of large LMs (Belrose et al., 2023; Gurnee et al., 2023; Bills et al., 2023).

Causality-based Interpretability. Causal mediation analysis is an important tool that is used to effectively attribute the causal effect of mediators on an outcome variable (Pearl, 2001). This paradigm was applied to investigate LMs by Vig et al. (2020), who proposed a framework based on causal mediation analysis to investigate gender bias. Variants of this approach were later applied to mechanistically interpret the inner workings of pre-trained LMs on other tasks such as subject-verb agreement (Finlayson et al., 2021), natural language inference (Geiger et al., 2021), indirect object identification (Wang et al., 2022), and to study their retention of factual knowledge (Meng et al., 2022).

Math and Arithmetic Reasoning. A growing body of work has proposed methods to analyze the performance and robustness of large LMs on tasks involving mathematical reasoning (Pal and Baral, 2021; Piękos et al., 2021; Razeghi et al., 2022; Cobbe et al., 2021; Mishra et al., 2022). In this area, Stolfo et al. (2023) use a causally-grounded approach to quantify the robustness of large LMs. However, the proposed formulation is limited to behavioral investigation with no insights into the models' inner mechanisms. To the best of our knowledge, our study represents the first attempt to connect the area of mechanistic interpretability to the investigation of the mathematical reasoning abilities in Transformer-based LMs.

3 Methodology

3.1 Background and Task

We denote an autoregressive language model as $\mathcal{G} : \mathcal{X} \rightarrow \mathcal{P}$. The model operates over a vocabulary V and takes a token sequence $x = [x_1, ..., x_T] \in \mathcal{X}$, where each $x_i \in V$. $\mathcal{G}$ generates a probability distribution $\mathbb{P} \in \mathcal{P} : \mathbb{R}^{|V|} \rightarrow [0, 1]$ that predicts possible next tokens following the sequence x. In this work, we study decoder-only Transformer-based models (Vaswani et al., 2017). Specifically, we focus on models that represent a slight variation of

the standard GPT paradigm, as they utilize parallel attention (Wang and Komatsuzaki, 2021) and rotary positional encodings (Su et al., 2022). The internal computation of the model's hidden states $h_t^{(l)}$ at position $t \in \{1, \ldots, T\}$ of the input sequence is carried out as follows:

$$\begin{aligned} h_t^{(l)} &= h_t^{(l-1)} + a_t^{(l)} + m_t^{(l)} \\ a_t^{(l)} &= \mathrm{A}^{(l)}\left(h_1^{(l-1)}, \ldots, h_t^{(l-1)}\right) \\ m_t^{(l)} &= W_{\text{proj}}^{(l)}\, \sigma\left(W_{fc}^{(l)}\, h_t^{(l-1)}\right) \\ &=: \mathrm{MLP}^{(l)}(h_t^{(l-1)}), \end{aligned} \tag{1}$$

where at layer l, σ is the sigmoid nonlinearity, $W_{fc}^{(l)}$ and $W_{\text{proj}}^{(l)}$ are two matrices that parameterize the multilayer perceptron (MLP) of the Transformer block and $\mathrm{A}^{(l)}$ is the attention mechanism.[2]

We consider the task of computing the result of arithmetic operations. Each arithmetic query consists of a list of operands $N = (n_1, n_2, \ldots)$ and a function f_O representing the application of a set of arithmetic operators $(+, -, \times, \div)$. We denote as $r = f_O(N)$ the result obtained by applying the operators to the operands. Each query is rendered as a natural language question through a prompt $p(N, f_O) \in \mathcal{X}$ such as *"How much is n_1 plus n_2?"* (in this case, $f_O(n_1, n_2) = n_1 + n_2$). The prompt is then fed to the language model to produce a probability distribution $\mathbb{P}$ over V. Our aim is to investigate whether certain hidden state variables are more important than others during the process of computing the result r.

3.2 Experimental Procedure

We see the model $\mathcal{G}$ as a causal graph (Pearl, 2009), framing internal model components, such as specific neurons, as mediators positioned along the causal path connecting model inputs and outputs. Following a causal mediation analysis procedure, we then quantify the contribution of particular model components by intervening on their activation values and measuring the change in the model's output. Previous work has isolated the effect of every single neuron within a model (Vig et al., 2020; Finlayson et al., 2021). However, this approach becomes impractical for models with billions of parameters. Therefore, for our main experiments, the elements that we consider as variables along the causal path described by the model are

[2]For brevity, layer normalization (Ba et al., 2016) is omitted as it is not essential for our analysis.

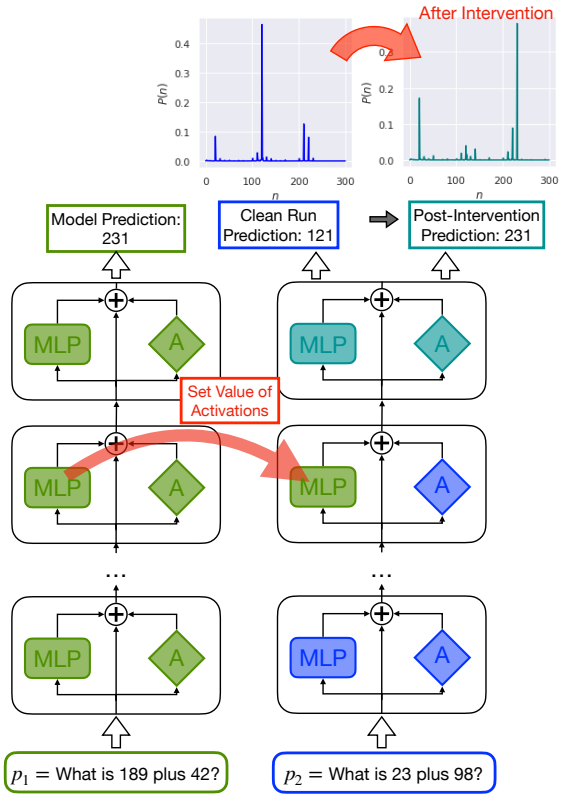

Figure 2: By intervening on the activation values of specific components within a language model and computing the corresponding effects, we identify the subset of parameters responsible for specific predictions.

the outputs of the $\mathrm{MLP}^{(l)}$ and $\mathrm{A}^{(l)}$ functions at each token t, i.e., $m_t^{(l)}$ and $a_t^{(l)}$.

To quantify the importance of modules $\mathrm{MLP}^{(l)}$ and $\mathrm{A}^{(l)}$ in mediating the model's predictions at position t, we use the following procedure.

1. Given f_O, we sample two sets of operands N, N', and we obtain $r = f_O(N)$ and $r' = f_O(N')$. Then, two input questions with only the operands differing, $p_1 = p(N, f_O)$ and $p_2 = p(N', f_O)$, are passed through the model.

2. During the forward pass with input p_1, we store the activation values $\bar{m}_t^{(l)} := \mathrm{MLP}^{(l)}(h_t^{(l-1)})$, and $\bar{a}_t^{(l)} := \mathrm{A}^{(l)}(h_1^{(l-1)}, \ldots, h_t^{(l-1)})$.

3. We perform an additional forward pass using p_2, but this time we *intervene* on components $\mathrm{MLP}^{(l)}$ and $\mathrm{A}^{(l)}$ at position t, setting their activation values to $\bar{m}_t^{(l)}$, and $\bar{a}_t^{(l)}$, respectively. This process is illustrated in Figure 2.

4. We measure the causal effect of the intervention on variables $m_t^{(l)}$ and $a_t^{(l)}$ on the model's prediction by computing the change in the probability values assigned to the results r and r'.